

Problem Set 3: Demand, Supply and Competitive Markets

Questions and Solutions

1) Ann, Bill and John's Willingness to Pay (WTP) for cups of coffee are given in table 1.

Q	Ann's WTP	Bill's WTP	John's WTP
1	2.7	2.5	4.3
2	5.1	4.5	7.9
3	7.2	6	10.8
4	9	7	13
5	10.5	7.5	14.5
6	11.7	7.5	15.3

Table 1: WTP for Ann, Bill and John

1a) If each cup of coffee costs \$2, how many cups of coffee does each of them buy?

Solution: The first step is to calculate the marginal willingness to pay for each of them which is given in table 2

Q	Ann's MWTP	Bill's MWTP	John's MWTP
1	2.7	2.5	4.3
2	2.4	2	3.6
3	2.1	1.5	2.9
4	1.8	1	2.2
5	1.5	0.5	1.5
6	1.2	0	0.8

Table 2: MWTP for Ann, Bill and John

Assuming that they buy an extra cup of coffee as long as their $MWTP \geq$ price of a cup, **Ann will buy 3 cups, Bill will buy 2 cups and John will buy 4 cups.**

Notice here that when we have a table, we can only work with integer units of the good. In that case MWTP can be $>$ price and the person will stop because going to the next integer value makes $MWTP <$ price. If we had equations, we could increase the quantity by tiny increments till $MWTP =$ price.

Notice also that we always assume that when the consumer is indifferent, i.e. $MWTP =$ price, they buy the extra unit. I will not explicitly state this on a test, I will assume you know it.

1b) If the price of a cup of coffee is \$2.4 what is the total demand from all three of them (i.e. market demand if we had only 3 of them on the demand side of the market)?

Solution: Following the logic above, we first figure out how much each of them will buy at a price of \$2.4 and sum it up to get total (market) demand. **We get 2 cups from Ann, 1 cup from Bill and 3 cups from John, a total of 6 cups of coffee demanded.**

Another way to see it is to order everyone's MWTP from highest to lowest and then count the total number of cups for which $MWTP \geq \$2.4$.

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
MWTP	4.3	3.6	2.9	2.7	2.5	2.4	2.2	2.1	2	1.8	1.5	1.5	1.5	1.2	1	0.8	0.5	0

Q	Deb's TC	Ebert's TC	Frank's TC
1	0.8	1.4	2.2
2	2.4	3.2	4.6
3	4.8	5.4	7.2
4	8	8	10
5	12	11	13
6	16.8	14.4	16.2

Table 3: Deb, Ebert and Frank's TCs of production

Q	Deb's MC	Ebert's MC	Frank's MC
1	0.8	1.4	2.2
2	1.6	1.8	2.4
3	2.4	2.2	2.6
4	3.2	2.6	2.8
5	4	3	3
6	4.8	3.4	3.2

Table 4: Deb, Ebert and Frank's MCs of production

We can see that when market price is \$2.4, 6 cups will be demanded in total. It is useful to think about it this way because then you can clearly link individual demand to market demand.

Notice that from a market perspective we don't care who buys the third cup. It is the person with the next highest MWTP who will buy the extra cup when the price drops.

1c) If we want a total of 10 cups of coffee demanded, what is the highest price coffee can be?

Solution: Following the same logic as the question above and looking at the ordered MWTPs, if price is above \$1.8 we will get fewer than 10 cups of coffee demanded. **The highest price is therefore \$1.8, the MWTP of the 10th cup**, at which we get exactly 10 cups (across the three people) for which $MWTP \geq 1.8$.

Who will buy these cups? To determine that, look back at the individual MWTPs: **Ann buys 4, Bill buys 2 and John buys the remaining 4 cups.**

2) Deb, Ebert and Frank are producing coffee and their total costs of producing given quantities are listed in table 3

2a) If the price of a cup of coffee is \$2, how many cups of coffee do each of them supply?

Solution: The first step is to calculate the marginal cost for each of them, which is given in table 4.

Assuming that they sell an extra cup of coffee as long as $MC \leq \text{price}$, **Deb will sell 2 cups, Ebert will sell 2 cups and Frank will sell no cups.**

Notice again that when we have a table, we can only work with integer units of the good. In that case MC can be $<$ price and the person will stop because going to the next integer value makes the $MC >$ price. If we had equations, we could increase the quantity by tiny increments till $MC = \text{price}$.

Notice also that we always assume that when the supplier is indifferent, i.e. $MC = \text{price}$, they sell the extra unit. I will not explicitly state this on a test, I will assume you know it.

2b) If the price of a cup of coffee is \$2.4 what is the total supply from all three of them (i.e. market supply if we had only 3 of them on the supply side of the market)?

Solution: Following the logic above, we first figure out how much each of them will sell at a price of \$2.4 and sum it up to get total (market) supply. **We get 3 cups from Deb, 3 cups from Ebert and 2 cups from Frank, a total of 8 cups of coffee supplied.**

Another way to see it is to order everyone's MC from lowest to highest and then count the total number of cups for which $MC \leq \$2.4$.

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
MC	0.8	1.4	1.6	1.8	2.2	2.2	2.4	2.4	2.6	2.6	2.8	3	3	3.2	3.2	3.4	4	4.8

We can see that when market price is \$2.4, 8 cups will be supplied in total.

2c) If we want a total of 10 cups of coffee supplied, what is the lowest price coffee can be?

Solution: Following the same logic as the question above and using the table of MCs arranged in increasing order, if price is below \$2.6 we will get fewer than 10 cups of coffee supplied. **The lowest price is therefore \$2.6, the MC of the 10th cup.** If price is = 2.6 then we get exactly 10 cups for which the price $\geq$ MC.

Who will sell these cups? To determine that, look back at the individual MCs: **Deb sells 3, Ebert sells 4 and Frank sells the remaining 3 cups.**

3) A perfectly competitive market consists of 15 individual buyers and 10 individual sellers. Each buyer's demand curve is given by $p = 20 - \frac{1}{5}q^D$. Each supplier's supply curve is given by $p = \frac{1}{5}q^S - 2$. (Notation: I'm using small letters for individuals and capital letters for markets to make things easier to keep track of visually. There is no deeper meaning to it.)

3a) If the market price is \$10, what is each individual buyer's quantity demanded and what is market demand?

Solution: To find the quantity demanded at a given price, we rewrite the equation as quantity as a function of price. If we don't, we end up summing prices, which is conceptually wrong. Our goal is to sum up individual quantity demands to get market quantity demanded.

$$\begin{aligned} p &= 20 - \frac{1}{5}q^D \\ \frac{1}{5}q^D &= 20 - p \\ q^D &= 100 - 5p \end{aligned}$$

Substitute $p = \$10$ into the demand equation to get each individual's demand as $q^D = 50$.

To get market demand we need to sum up individual demands at a given price. If we have 15 buyers each demanding 50 units, then **market demand at a price of \$10 is $50 * 15 = 750$.**

3b) If market price is \$10, what is the MWTP for the last unit bought?

Solution: We know that for every unit bought it was bought because $MWTP \geq$ price. When dealing with equations, we can say something specific about the last unit bought. For the last unit, the reason we stopped at that unit and didn't go further is because at that unit $MWTP =$ price. Increasing quantity would cause $MWTP <$ price. So **if we are talking about the last unit bought when the price is \$10, we know the MWTP for that unit is \$10.**

Question for you, does it matter if we look at the market demand curve or the individual demand curve to identify the MWTP? Hint: See question 2 above.

3c) At what price is market demand $Q^D = 500$?

Solution: We can derive a function for market demand now that will be useful when solving for market equilibrium later. Using the same logic that market demand (Q^D) is the sum of individual demand ($q^D(p)$) at price p we have

$$Q^D = \sum_{i=1}^{15} q_i^D(p) = 15 * (100 - 5p) = 1500 - 75p$$

At this point you should make sure you understand why we can't sum up the inverse demand curves (hint: that would be a vertical summation, not a horizontal summation, on a graph).

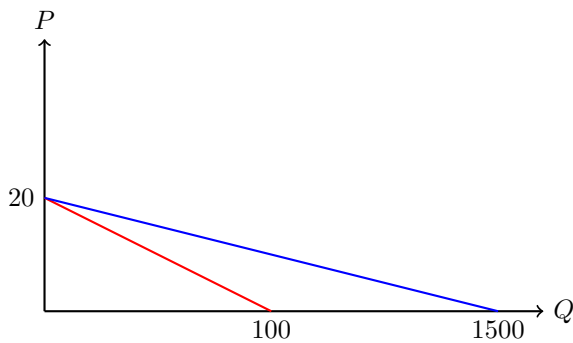


Figure 1: Individual (red) and Market (blue) demand curves

To check what price makes market demand equal to 500 we just substitute $Q^D = 500$ into our market demand equation and solve for p

$$\begin{aligned} Q^D &= 1500 - 75p \\ 500 &= 1500 - 75p \\ 75p &= 1500 - 500 \\ p &= \frac{1000}{75} = 13.33 \end{aligned}$$

So the price at which market demand is 500 is \$13.33. To double-check, work out what individual demand is when $p = 13.33$. Plugging $p = 13.33$ into $q^D(p) = 100 - 5 * 13.33 = 33.35$. With 15 people on the demand side, total market demand is $15 * 33.35 = 500.25$ which with rounding is 500. This is a good way on the exam to make sure you've correctly summed up individual demand to get market demand.

3d) Draw each individual's demand curve and the market demand curve on a clearly labeled diagram.

Solution: Find your intercepts on the individual and market demand curves by plugging in $p = 0$ to get the Q axis intercept and $q^D, Q^D = 0$ to get the P axis intercept. Putting these numbers onto a graph gives us figure 1.

Do you know why the slopes of the two curves are different? Think about what happens when the price drops by \$1: how does individual demand change, and how does market demand change? Why are they different? Can you use the numbers to guide your intuitive explanation?

3e) If the market price is \$10, what is each individual supplier's quantity supplied and what is market supply?

Solution: As with demand, the inverse supply curve just expresses price as a function of quantity instead of the other way round. To find the quantity supplied at a given price, we convert the inverse supply curve into a supply curve (i.e. a quantity as a function of price equation)

$$\begin{aligned} p &= \frac{1}{5}q^S - 2 \\ \frac{1}{5}q^S &= p + 2 \\ q^S &= 10 + 5p \end{aligned}$$

Substitute $p = \$10$ into the supply equation to get each individual's supply as $q^S = 60$.

To get market supply we need to sum up individual supplies at a given price. If we have 10 sellers each selling 60 units, then **market supply at a price of \$10 is $60 * 10 = 600$.**

3f) At what price is market supply $Q^S = 500$?

Solution: We can derive a function for market supply now that will be useful when solving for market equilibrium later. Using the same logic in the market demand derivation we have

$$Q^S = \sum_{i=1}^{10} q_i^S(p) = 10 * (10 + 5p) = 100 + 50p$$

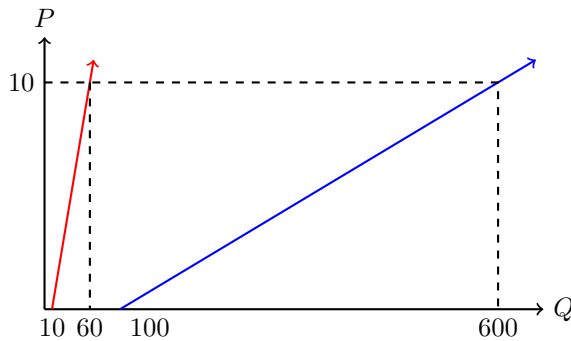


Figure 2: Individual (red) and Market (blue) supply curves

At this point you should make sure you understand why we can't sum up the inverse supply curves (hint: that would be a vertical summation, not a horizontal summation, on a graph).

To check what price makes market supply equal to 500 we just substitute $Q^S = 500$ into our market supply equation and solve for p

$$\begin{aligned} Q^S &= 100 + 50p \\ 500 &= 100 + 50p \\ 50p &= 500 - 100 \\ p &= \frac{400}{50} = 8 \end{aligned}$$

So the price at which market supply is 500 is \$8. To double-check, work out what individual supply is when $p = 8$. Plugging $p = 8$ into $q^S(p) = 10 + 5 * 8 = 50$. With 10 people on the supply side, total market supply is $10 * 50 = 500$. Again, this is a good way on the exam to make sure you've correctly summed up individual supply to get market supply.

3g) Draw each individual's supply curve and the market supply curve on a clearly labeled diagram.

Solution: Find your intercepts on the individual and market supply curves by plugging in $p = 0$ to get the Q axis intercepts for both curves. Then notice that as price increases supply increases, and price can't be negative, so we don't have a P axis intercept. The supply curves are given in figure 2.

Note: I've chosen my numbers so that at a price of zero, suppliers are still willing to supply goods. Is this realistic? No. I've done it to catch any instinct to draw the supply curves you've seen in the text and make you think about what exactly the intercepts mean.

Do you know why the slopes of the two curves are different? Think about what happens when the price increases by \$1, how does individual supply change, and how does market supply change? Why are they different? Can you use the numbers to guide your intuitive explanation?

3h) What is market price?

Solution: Market price sets market quantity demanded equal to market quantity supplied

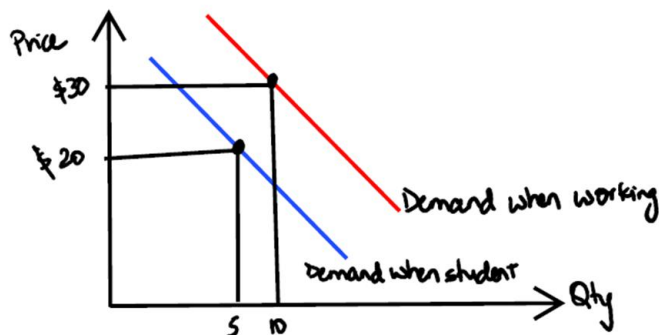
$$\begin{aligned} Q^D &= Q^S \\ 1500 - 75p &= 100 + 50p \\ 1500 - 100 &= 75p + 50p \\ p &= \frac{1400}{125} = 11.2 \end{aligned}$$

The market price is \$11.20, where market quantity demanded equals market quantity supplied. From the numbers in the earlier parts, can you explain why the price has to be above \$10?

4) True/False/Uncertain. We are looking at Jada's clothing choice. When she was a student, Jada bought 5 T-shirts when the price was \$20 a T-shirt. The next time we observe Jada buying the same T-shirts, she is working.

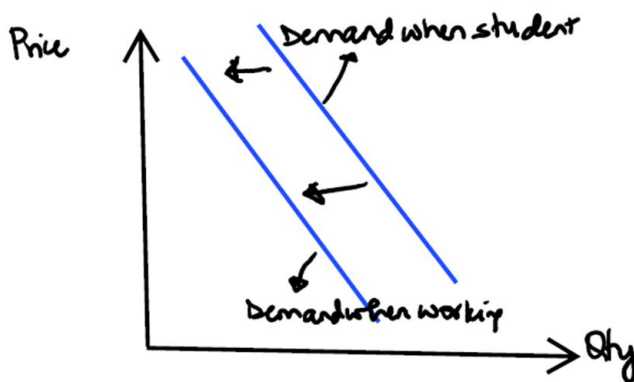
The price is \$30 (inflation adjusted so comparable) a T-shirt and she buys 10 T-shirts. Statement: Based on these two observations, Jada's demand curve for T-shirts is upward sloping. True, False or Uncertain? Explain.

Solution: Uncertain. The thought experiment to derive Jada's demand curve needs to hold all else constant and vary only the price of T-shirts. Here there could be an income difference between the two observations (student vs. working). **Jada could have a downward sloping demand curve, but because of the shift in income we are picking up points on two different demand curves, which makes it look like upward sloping demand even when it is not.**



5) True/False/Uncertain. Ramen noodles are an inferior good for Houman. Statement: When Houman graduates from college and becomes an economist, his demand for Ramen increases. True, False or Uncertain? Explain.

Solution: False. When Houman becomes an economist, his income is higher compared to when he was a student. For an inferior good, when the income rises, the quantity demanded at the same price falls. **The entire demand curve shifts in, so his demand for Ramen decreases.**



6) Assume that all the markets below are perfectly competitive markets.

This would be a good time, for all the sample markets below, to think about whether you can treat buyers and sellers as price takers in that market. I don't expect you to know the answer, and we're just going to assume it here. I just want you to think about the assumption in the context of real examples. It will help you understand exactly what that assumption really means which will be helpful later in the course.

For each change below, tell us what happens to market price and market quantity. You can assume typical downward sloping demand and upward sloping supply curves.

6a) The ice-cream market: The price of vanilla beans increases dramatically. (Vanilla (from beans) is a main ingredient in icecream)

Solution: Increasing the price of vanilla shifts up (and in) the icecream supply curve, **raising the price of icecream and decreasing the quantity of icecream.**

6b) The vodka market: A decrease in the production of vanilla extract. (Vodka is a main ingredient in vanilla extract production, along with vanilla beans.)

Solution: The demand for vodka shifts in, lowering price and quantity sold.

6c) The market for imitation vanilla extract: The price of vanilla beans increases. (Vanilla extract is made from wood pulp, not the vanilla bean)

Solution: It is reasonable to assume that imitation vanilla and real vanilla extract are substitutes. Increasing the price of vanilla beans shifts in the real vanilla extract supply curve, raising the price of real vanilla extract. This shifts out the demand curve for imitation vanilla extract. The shift out of the demand curve means we move up along the supply curve, so **the price and quantity transacted of imitation vanilla extract both increase.**

6d) The printer market: The price of printer ink cartridges decreases.

Solution: It is reasonable to assume printers and cartridges are complements. This shifts out the demand curve for printers, so **price and quantity increase as we move up along the supply curve.** (One way to think about it: using a printer requires an ink cartridge. A decrease in cartridge price makes printing less expensive, thus increasing the quantity demanded of printers at the same printer price.)

6e) The market for physical books: Amazon reduces the prices of kindle e-books.

Solution: A decrease in the price of a substitute will shift the demand curve in. **As we move down the supply curve, both price and quantity fall.**

6f) Vanilla bean market: The vanilla bean crop in Madagascar (major producer of vanilla beans) failed.

Solution: The supply curve shifts in. **We move up the demand curve, so price increases and quantity decreases.**

6g) Fast food : Canada falls in a recession and people lose their jobs. At the same time, new technology allows fast food outlets to replace people with machines and reduce the cost of producing fast food. (Fast food is an inferior good).

Solution: Losing a job means lower incomes and since fast food is inferior, this shifts the demand curve out. A decrease in production costs, shifts supply out. **Both of these increase the market quantity of fast food, but have an ambiguous effect on its price.**

6h) Ontario kale: An increased awareness on the health benefits of kale makes more people buy kale. Changing climate has made Ontario land less suitable to growing kale.

Solution: More people buying kale shifts demand out. Less suitable land increases the opportunity cost of producing kale, which decreases the number of farmers producing it (relative to the alternatives), so supply shifts in. **Both factors increase the price of kale, but the effect on the market quantity of Ontario kale is ambiguous.**

6i) Elastic: COVID-19 causes an increase in the demand for cloth face masks.

Solution: Elastic is one component of a cloth face mask, which makes it a complementary input in the production of face masks. **This will increase the demand for elastic and raise both its price and the quantity sold.** Note: you could also make an argument that COVID-19 has reduced the supply of elastic because of factory closures and disrupted global supply chains. See [Toilet paper, cleaning products and now, elastic?](#) by Jessica Villagomez. Chicago Tribune, April 25, 2020.

7) Assume that the market for take-out meals is perfectly competitive. For each change below, tell us whether it is a shift of the curve (i.e. of the demand or supply curve) or a movement along the curve.

COVID-19 has affected the market for take-out meals around the UofT campus. Let's think of some ways using the examples below.

For each, focus just on that one change, assuming everything else held constant at the Fall 2019 levels.

7a) Fewer students on campus.

Solution: Individual demand is unaffected, but there are fewer individuals in the market, so **market demand shifts in.** For the same price, the total quantity demanded is lower.

7b) Students lose their part-time jobs.

Solution: Income is now lower. **Demand shifts, but whether it shifts in or out depends on whether meals are inferior or normal; if we assume they are normal, demand shifts in.** For the same price of a meal, the quantity demanded decreases or for the same quantity of meals, the Marginal Willingness to Pay (MWTP) is lower than before. The opposite is true if inferior.

7c) Students prefer to eat at home because they're worried about getting infected if they eat food cooked by someone else.

Solution: Preferences have changed, so **demand shifts in.** For the same price of a meal, the quantity demanded decreases or for the same quantity of meals, the Marginal Willingness to Pay (MWTP) is lower than before.

7d) Groceries are much more expensive because fresh food prices have risen.

Solution: Demand side: the price of a related good has changed, so **demand shifts out.** For the same price of a meal, the quantity demanded increases, as cooking at home, a substitute, is now more expensive. Or for the same quantity of meals, the Marginal Willingness to Pay (MWTP) is higher than before.

Supply side: the price of an input has changed, so **supply shifts in.** For the same price of a meal, the quantity supplied decreases; or for the same quantity of meals, the MC is higher than before.

7e) The cost of making an extra meal increases due to additional sanitary measures restaurants have to take.

Solution: The cost of an input has changed, so **supply shifts in.** For the same price of a meal, the quantity supplied decreases; or for the same quantity of meals, the MC is higher than before.

7f) The price of a restaurant meal increases.

Solution: Finally, **a movement along the curve!** All else held equal, a change in price is a movement along the demand or the supply curve.

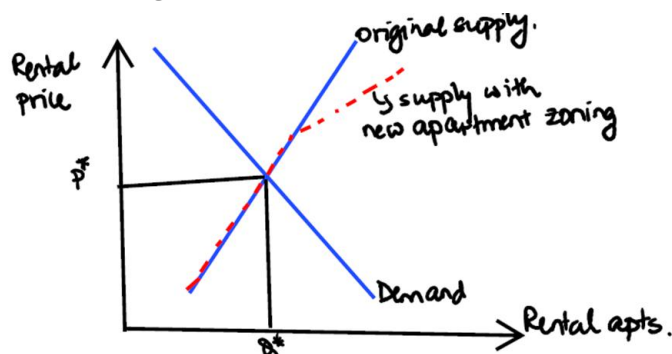
8) Apartment rents in Toronto are **very high**. You may know this from personal experience. Assume that rents are determined in a market based on the demand and supply of rental units.

You are at a recent community town hall where residents are asking the city to change zoning laws to allow more rental apartments to be built downtown. They hope this will increase the supply of apartments and decrease rents.

One person says that changing zoning laws this way will not solve the problem of high rents. Their argument is that building apartments is expensive given the cost of land and construction. They say that the only rental apartments that will be constructed will cater to very high-income renters.

Can you explain how to graph this argument? How can it be that an increase in supply will not change market price?

Solution: Here you want to think about what determines the supply curve. A supplier will produce the extra unit as long as their benefit (rental price) is greater than their marginal cost (MC) of production. Suppose that, due to high costs of production, every additional unit has an MC higher than the current rental price of apartments. Even if you allowed builders to build apartments, they will only be willing to supply the additional apartments if the price is even higher than the current market equilibrium. **Supply will shift out, but only above the current equilibrium. Since that leaves the current equilibrium unchanged, rents will remain high.**



The city of Toronto is trying to address the problem by changing the zoning laws in a different way—encouraging laneway rental units. Can you explain why this could increase supply in a way that is different from the argument above? See “[Laneway housing begins to take shape under Toronto’s new rules](#)” by Emily Mathieu in The Toronto Star. March 18, 2019.

- 9) The Market for Kidneys. We can’t buy and sell organs. However, there are two sides of the market—people who need kidneys and those who would be willing to give up one kidney to a person in need (we can survive just fine with only one). Since there is no market to buy and sell kidneys, can you design a mechanism to help people make this trade? What would it look like?

Solution: The purpose of this question is to get you to think about the role of the market and how prices determine allocations. If we have two sides of the market, is there another mechanism, more effective and requiring minimal outside intervention, to allocate goods to the people with the highest value (highest MWTP)? At the same time you need to identify people willing to give up the good and match them together to make a trade that makes both sides better off.

Check out Planet Money’s [Episode 372: How Do You Decide Who Gets Lungs?](#)¹ It deals with this exact dilemma and shows how there could be unexpected outcomes from any allocation mechanism.

Checked with Claude (Anthropic) for grammar, clarity and errors. Tell me if you find one I missed.

1

Planet Money [Episode 372: How Do You Decide Who Gets Lungs?](#). April 29, 2015. Available at: <https://www.npr.org/sections/money/2015/04/29/403114496/episode-372-how-do-you-decide-who-gets-lungs>